

# PuzzleForge — Project Summary

**Project:** PuzzleForge — Agentic Puzzle Game Design and Generation System

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**Track:** A (Technical Build)

**Course:** 94-815 Agentic Systems Studio, CMU Heinz College, Spring 2026

## Summary

PuzzleForge is a multi-agent system that transforms a user's free-text game concept into a playable, browser-based Sokoban puzzle game. The Game Designer agent chooses between push and slide mechanics based on the user's concept and designs a complete visual theme (emoji icons, color scheme, entity names). The system uses 5 coordinated agents across two layers (Design and Implementation) orchestrated by a LangGraph StateGraph pipeline. Every generated level is verified solvable by a BFS solver with deadlock detection — solvability is never determined by LLM judgment. The Developer agent classifies QA failures using 10 deterministic routing criteria and dispatches them to either the Debugger (config bugs) or Level Designer (design flaws) for iterative repair, capped at 3 debug cycles.

## Key Results

- **82% solvability rate** across 4 live pipeline runs (14/17 levels; push: 90%, slide: 71%)
- 100% pipeline completion rate (4/4 runs)
- Game Designer correctly chose mechanic in all 4 runs (push vs slide)
- Mean 6,448 tokens per run (well within 80K budget)
- Mean 30.9 seconds per run (well within 5-minute timeout)
- 8 documented failures with root cause analysis and fixes
- 11 test cases executed, all passing
- **Baseline comparison:** Single-LLM achieves 65% solvability (slide: 30%); multi-agent pipeline achieves 82% (slide: 71%) — debug loop more than doubles slide solvability

## Repository Link

(Insert GitHub repository URL here)

## Video Link

(Insert 5-minute video URL here)

## Architecture Diagram

See docs/architecture\_diagram.pdf

## **Final Report**

See docs/final\_report.pdf